

Kush Jalan

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EDUCATION

Mahindra University

B.Tech Computer Science

Obul Reddy College

CBSE – Class XII

Aug 2023 - Present

Current GPA: 9.4

June 2021 – Feb 2023

COURSEWORK

Courses: Object-Oriented Programming, Data Structures & Algorithms, Operating System, Database Management, Machine Learning, Software Engineering, Social Computing, Quantum Computing, Gen Ai and LLM.

Awards: Merit Scholarship from Mahindra University for ranking in the top 10% of the batch in the years 2023, 2024 and 2025.

SKILLS

Languages: Java, Python, C, C++, JavaScript

Technologies: REST APIs, Git, GitHub, Linux (Basics), HTML, CSS

Database: MySQL, PostgreSQL

Libraries: Pandas, NumPy, Matplotlib, PyTorch, Scikit-learn

Tools: VS Code, CI/CD (GitHub Actions), Agile (Scrum basics), Docker

PROJECTS

Low-Latency Order Matching Engine | *C++, STL, Multithreading*

Feb 2026

- Built a high-performance order matching engine using price-time priority with $O(\log n)$ order matching for limit and market orders
- Implemented concurrent order processing using multithreading and mutex locks, with latency benchmarking and logging
- Containerized the application using Docker for consistent deployment and reproducible testing environments

Interactive Voice Assistant | *Python, SpeechRecognition, gTTS/pyttsx3, NLP, APIs*

Dec 2024

- Designed and developed a voice assistant system enabling real-time command execution using speech recognition, NLP, and external APIs.
- Implemented speech recognition pipelines to process user commands and generate conversational responses.
- Integrated NLP and audio processing modules to enable seamless hands-free communication.

Predictive Maintenance of Turbofan Engine | *python, NumPy, Pandas, Random Forest, XGBoost*

Mar 2025

- Developed a predictive maintenance system using CMAPSS Dataset to estimate Remaining Useful Life (RUL) and classify all engine states with an accuracy of 87 percent.
- Implemented ensemble learning techniques including Random Forest and XGBoost.

Racism and Hate Speech Detector | *Python, Pytorch, Hugging Face Transformers, BERT, LSTM*

Oct 2025

- Developed a Racism and Hate speech detection system using NLP.
- Implemented NLP pipeline with tokenization and deep learning models including LSTM and BERT using Hugging Face Transformers.
- Achieved 95 percent classification accuracy on test dataset using BERT fine-tuning.

INTERESTS/EXTRA-CURRICULAR ACTIVITIES

Interests: Chess, Basketball, Anime, Coding, Video Games

Secured Second Place in an Inter-School Basketball Tournament.

Volunteer: Social service initiatives at Mahindra University

Actively volunteered in multiple social service initiatives and university events at Mahindra University.